

COMM114: Generative AI - Guided Diffusion Models

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November 2025

Recap from previous lectures

- Let PDF of $\mathbf{X}_0$ be $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e., $p_{\mathbf{X}_0}(\mathbf{x}_0) = p_{init}(\mathbf{x})$)
- Let PDF of $\mathbf{X}_1$ be $p_{data}(\mathbf{x})$ (i.e., $p_{\mathbf{X}_1}(\mathbf{x}_1) = p_{data}(\mathbf{x})$)
- Define $\mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1$ where α_t and β_t satisfy:
 - α_t and β_t are scalar differentiable functions over $t \in [0, 1]$.
 - When $t = 0$, $\alpha_{t=0} = 0$ and $\beta_{t=0} = 1$.
 - When $t = 1$, $\alpha_{t=1} = 1$ and $\beta_{t=1} = 0$.
- Consider the following ODE

$$\frac{d\mathbf{X}_t}{dt} = \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d \beta_t^2}{dt} \right) \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t | \mathbf{X}_1) + \frac{d \log \alpha_t}{dt} \mathbf{X}_t \quad (1)$$

- It can be shown that $\mathbf{X}_t$, the solution to the above ODE, transforms the distribution $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ at $t = 0$ to the data distribution p_{data} at $t = 1$.
- In other words, the probability path of $\mathbf{X}_t$ satisfy the boundary probability conditions.

Recap from previous lectures (cont.)

Prior knowledge:

It is known that the conditional distribution $p_{t|1}(\mathbf{x}_t|\mathbf{x}_1)$ is Gaussian. As a result, the score function $\nabla_{\mathbf{x}_t} \log p_{t|1}(\mathbf{x}_t|\mathbf{x}_1)$ in (4) takes an alternative form:

$$\nabla_{\mathbf{x}_t} \log p_{t|1}(\mathbf{x}_t|\mathbf{x}_1) = -\frac{\mathbf{x}_t - \alpha_t \mathbf{x}_1}{\beta_t^2} = -\frac{\mathbf{x}_0}{\beta_t}$$

where $\mathbf{x}_0$ is the Gaussian noise when constructing $\mathbf{x}_t$.

- **Input:** data samples $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^n$.
- **Assumption:** Assume $\{\mathbf{x}_i\}_{i=1}^n$ are drawn from the data distribution $p_{data}(\cdot)$.
- A neural network $\hat{\epsilon}_t^\theta(\cdot)$ is trained by optimizing

$$\begin{aligned} & \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}[0,1], \mathbf{x} \sim p_{data}, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} \left[\left\| \hat{\epsilon}_t^\theta(\mathbf{x}_t) - \epsilon \right\|^2 \right] \quad [\epsilon \text{ corresponds to } \mathbf{x}_0] \\ &= \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}[0,1], \mathbf{x} \sim p_{data}, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} \left[\left\| \hat{\epsilon}_t^\theta(\alpha_t \mathbf{x} + \beta_t \epsilon) - \epsilon \right\|^2 \right] \end{aligned} \quad (2)$$

Recap from previous lectures (cont.)

- Once the neural network $\hat{\epsilon}_t^\theta(\cdot)$ is trained, the ODE (4) can be approximated in terms of $\hat{\epsilon}_t^\theta(\cdot)$ as:

$$\begin{aligned} \frac{d\mathbf{x}_t}{dt} &= \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d\beta_t^2}{dt} \right) \nabla_{\mathbf{x}_t} \log p_{t|1}(\mathbf{x}_t | \mathbf{x}_1) + \frac{d \log \alpha_t}{dt} \mathbf{x}_t \\ \Leftrightarrow \frac{d\mathbf{x}_t}{dt} &= \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d\beta_t^2}{dt} \right) \frac{-\hat{\epsilon}_t^\theta(\mathbf{x}_t)}{\beta_t} + \frac{d \log \alpha_t}{dt} \mathbf{x}_t \\ \Leftrightarrow d\mathbf{x}_t &= (d\beta_t - \beta_t d \log \alpha_t) \hat{\epsilon}_t^\theta(\mathbf{x}_t) + d \log \alpha_t \mathbf{x}_t \text{ where } \mathbf{x}_0 \sim p_{init} \end{aligned} \quad (3)$$

Remark: With the neural network $\hat{\epsilon}_t^\theta(\cdot)$, new data samples can be generated by simulating the ODE (3) via Euler method.

Objectives of this lecture

- Guided diffusion models
- Classifier-free guidance for guided sampling

Guided Diffusion Models

- Examples: The additional input to the guided diffusion model could be labels of the images.



Guided FM (cont.)

- Examples: The additional input to the guided diffusion model could be the text prompts.



a massive alien space ship that is shaped like a pretzel.



A crab made of cheese on a plate



A kangaroo holding a beer, wearing ski goggles and passionately singing silly songs.



An entire universe inside a bottle sitting on the shelf at walmart on sale.



A cheesburger surfing the vibe wave at night

Changes from Diffusion Models to Guided Diffusion Models

- Suppose $p_{data}(\mathbf{x}, y)$ is a joint PDF of both the data sample and the additional information y .
- Take the additional information y as input to the related quantities in an unguided diffusion model.

diffusion model	guided diffusion model
$\left(\begin{array}{c} \text{target} \\ \text{score} \end{array} \right): \nabla_{\mathbf{x}_t} \log p_{t 1}(\mathbf{x}_t \mathbf{x}_1)$	$\left(\begin{array}{c} \text{guided target} \\ \text{score} \end{array} \right): \nabla_{\mathbf{x}_t} \log p_{t 1}(\mathbf{x}_t \mathbf{x}_1, \mathbf{y})$
$(\text{ model }): \hat{\epsilon}_t^{\theta}(\mathbf{x}_t)$	$\left(\begin{array}{c} \text{guided} \\ \text{model} \end{array} \right): \hat{\epsilon}_t^{\theta}(\mathbf{x}_t \mathbf{y})$
$\left(\begin{array}{c} \text{probability} \\ \text{path} \end{array} \right): p_{\mathbf{X}_t^{\theta}}(\mathbf{x}_t)$	$\left(\begin{array}{c} \text{guided} \\ \text{prob. path} \end{array} \right): p_{\mathbf{X}_t^{\theta} \mathbf{Y}}(\mathbf{x}_t \mathbf{y})$
$d\mathbf{x}_t = (d\beta_t - \beta_t d \log \alpha_t) \hat{\epsilon}_t^{\theta}(\mathbf{x}_t) + d \log \alpha_t \mathbf{x}_t$	$d\mathbf{x}_t = (d\beta_t - \beta_t d \log \alpha_t) \hat{\epsilon}_t^{\theta}(\mathbf{x}_t \mathbf{y}) + d \log \alpha_t \mathbf{x}_t$

Training a Guided Diffusion Model

- Let PDF of $\mathbf{X}_0$ be $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e., $p_{\mathbf{X}_0}(\mathbf{x}_0) = p_{init}(\mathbf{x})$)
- Let PDF of $(\mathbf{X}_1, \mathbf{Y})$ be $p_{data}(\mathbf{x}, \mathbf{y})$ (i.e., $p_{\mathbf{X}_1, \mathbf{Y}}(\mathbf{x}_1, \mathbf{y}) = p_{data}(\mathbf{x}, \mathbf{y})$)
- Define $\mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1$ where α_t and β_t satisfy:
 - α_t and β_t are scalar differentiable functions over $t \in [0, 1]$.
 - When $t = 0$, $\alpha_{t=0} = 0$ and $\beta_{t=0} = 1$.
 - When $t = 1$, $\alpha_{t=1} = 1$ and $\beta_{t=1} = 0$.
- Objective function for $\hat{\epsilon}_t^\theta(\mathbf{x}_t|\mathbf{y})$:

$$\begin{aligned} & \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}[0,1], (\mathbf{x}, \mathbf{y}) \sim p_{data}, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} \left[\left\| \hat{\epsilon}_t^\theta(\mathbf{x}_t|\mathbf{y}) - \epsilon \right\|^2 \right] \quad [\epsilon \text{ corresponds to } \mathbf{X}_0] \\ &= \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}[0,1], (\mathbf{x}, \mathbf{y}) \sim p_{data}, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} \left[\left\| \hat{\epsilon}_t^\theta(\alpha_t \mathbf{x} + \beta_t \epsilon | \mathbf{y}) - \epsilon \right\|^2 \right] \end{aligned}$$

Sampling from a guided Diffusion Model

- **Sample:** Generating new samples by solving the ODE with an **input y** :

$$d\mathbf{x}_t = (d\beta_t - \beta_t d\log \alpha_t) \hat{\epsilon}_t^\theta(\mathbf{x}_t|y) + d\log \alpha_t \mathbf{x}_t$$

Remark: One can generate many data samples for the same input y ,

Classifier-free guided Diffusion Model

- **Objective of classifier-free guidance:** To further improve the sampling quality and diversity of the generated data samples of the conventional guided Diffusion model.
- Basic Idea: Introduce a special symbol $\emptyset$ and let $y \in \mathcal{Y} \cup \{\emptyset\}$.
 - When $y = \emptyset$, we use $\hat{\epsilon}_t^\theta(x_t|y = \emptyset)$ to denote the **unguided noise predictor**.
 - When $y \in \mathcal{Y}$, we use $\hat{\epsilon}_t^\theta(x_t|y)$ to denote the guided noise predictor.
 - The expression $\hat{\epsilon}_t^\theta(x_t|y = \emptyset)$ implies the noise predictor is conditioned on **nothing**.
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Classifier-free guided Diffusion Model (cont.)

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 - When $y \in \mathcal{Y}$, we use $\hat{\epsilon}_t^\theta(\mathbf{x}_t|y)$ to denote the guided noise predictor.
 - The expression $\hat{\epsilon}_t^\theta(\mathbf{x}_t|y = \emptyset)$ implies the noise predictor is conditioned on **nothing**.
- We will use both $\hat{\epsilon}_t^\theta(\mathbf{x}_t|y)$ and $\hat{\epsilon}_t^\theta(\mathbf{x}_t|\emptyset)$ to do a better sampling.
 - $\mathbf{u}_t^\theta(\mathbf{x}_t|\emptyset)$ provides some additional information to help the sampling process.

Classifier-free guided Diffusion Model (cont.)

- Let PDF of $\mathbf{X}_0$ be $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e., $p_{\mathbf{X}_0}(\mathbf{x}_0) = p_{init}(\mathbf{x})$)
- Let PDF of $(\mathbf{X}_1, \mathbf{Y})$ be $p_{data}(\mathbf{x}, \mathbf{y})$ (i.e., $p_{\mathbf{X}_1, \mathbf{Y}}(\mathbf{x}_1, \mathbf{y}) = p_{data}(\mathbf{x}, \mathbf{y})$)
- Define $\mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1$ where α_t and β_t satisfy:
 - α_t and β_t are scalar differentiable functions over $t \in [0, 1]$.
 - When $t = 0$, $\alpha_{t=0} = 0$ and $\beta_{t=0} = 1$.
 - When $t = 1$, $\alpha_{t=1} = 1$ and $\beta_{t=1} = 0$.
- Revisit the following ODE

$$\boxed{\frac{d\mathbf{X}_t}{dt} = \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d \beta_t^2}{dt} \right) \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t | \mathbf{X}_1, \mathbf{Y}) + \frac{d \log \alpha_t}{dt} \mathbf{X}_t} \quad (4)$$

Objective of classifier-free guidance

- In the following, we are going to derive an alternative score function to replace $\nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t | \mathbf{X}_1, \mathbf{Y})$ to improve the sampling quality in the end.

Classifier-free guided Diffusion Model (cont.)

- By applying the Bayesian formula, we have

$$\begin{aligned} & \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t | \mathbf{X}_1, \mathbf{Y}) \\ &= \nabla_{\mathbf{X}_t} \log \frac{p_{t|1}(\mathbf{X}_t, \mathbf{Y} | \mathbf{X}_1)}{p(\mathbf{Y})} \\ &= \nabla_{\mathbf{X}_t} \log \frac{p_{t|1}(\mathbf{X}_t | \mathbf{X}_1) p(\mathbf{Y} | \mathbf{X}_t, \mathbf{X}_1)}{p(\mathbf{Y})} \\ &= \nabla_{\mathbf{X}_t} \log \underbrace{p_{t|1}(\mathbf{X}_t | \mathbf{X}_1)}_{\text{unguided conditional prob.}} + \nabla_{\mathbf{X}_t} \log p(\mathbf{Y} | \mathbf{X}_t, \mathbf{X}_1) \end{aligned}$$

Classifier-free guided Diffusion Model (cont.)

- We introduce an additional weighting factor $w > 0$

$$\nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t | \mathbf{X}_1) + w \nabla_{\mathbf{X}_t} \log p(Y | \mathbf{X}_t, \mathbf{X}_1) \quad (5)$$

- When $w > 1$, the contribution of $\nabla_{\mathbf{X}_t} p(Y | \mathbf{X}_t, \mathbf{X}_1)$ will be amplified, leading to better image sampling quality in the end.

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Classifier-free guided Diffusion Model (cont.)

- We introduce an additional weighting factor $w > 0$

$$\nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t|\mathbf{X}_1) + w \nabla_{\mathbf{X}_t} \log p(Y|\mathbf{X}_t, \mathbf{X}_1) \quad (6)$$

- When $w > 1$, the contribution of $\nabla_{\mathbf{X}_t} p(Y|\mathbf{X}_t, \mathbf{X}_1)$ will be amplified, leading to better image sampling quality in the end.
- Next we consider simplifying (6)

$$\begin{aligned} & \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t|\mathbf{X}_1) + w \nabla_{\mathbf{X}_t} \log p(Y|\mathbf{X}_t, \mathbf{X}_1) \\ &= \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t|\mathbf{X}_1) + w \nabla_{\mathbf{X}_t} \frac{\log p(\mathbf{X}_t, Y|\mathbf{X}_1)}{\log p_{t|1}(\mathbf{X}_t|\mathbf{X}_1)} \quad [\text{Bayesian formula}] \\ &= (1 - w) \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t|\mathbf{X}_1) + w \nabla_{\mathbf{X}_t} \log p(\mathbf{X}_t, Y|\mathbf{X}_1) \\ &= (1 - w) \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t|\mathbf{X}_1) + w \nabla_{\mathbf{X}_t} \log \left[p(\mathbf{X}_t|Y, \mathbf{X}_1) p(Y|\mathbf{X}_1) \right] \\ &= (1 - w) \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t|\mathbf{X}_1) + w \nabla_{\mathbf{X}_t} \log p(\mathbf{X}_t|Y, \mathbf{X}_1) \end{aligned}$$

Classifier-free guided Diffusion Model (cont.)

- Let PDF of $\mathbf{X}_0$ be $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e., $p_{\mathbf{X}_0}(\mathbf{x}_0) = p_{init}(\mathbf{x})$)
- Let PDF of $(\mathbf{X}_1, \mathbf{Y})$ be $p_{data}(\mathbf{x}, \mathbf{y})$ (i.e., $p_{\mathbf{X}_1, \mathbf{Y}}(\mathbf{x}_1, \mathbf{y}) = p_{data}(\mathbf{x}, \mathbf{y})$)
- Define $\mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1$ where α_t and β_t satisfy:
 - α_t and β_t are scalar differentiable functions over $t \in [0, 1]$.
 - When $t = 0$, $\alpha_{t=0} = 0$ and $\beta_{t=0} = 1$.
 - When $t = 1$, $\alpha_{t=1} = 1$ and $\beta_{t=1} = 0$.
- Consider the following ODE instead

$$\begin{aligned} \frac{d\mathbf{X}_t}{dt} = & \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d \beta_t^2}{dt} \right) \left[(1 - w) \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t | \mathbf{X}_1) + w \nabla_{\mathbf{X}_t} \log p(\mathbf{X}_t | Y, \mathbf{X}_1) \right] \\ & + \frac{d \log \alpha_t}{dt} \mathbf{X}_t \end{aligned} \quad (7)$$

- $\nabla_{\mathbf{X}_t} p(\mathbf{X}_t | Y, \mathbf{X}_1)$: guided score function
- $\nabla_{\mathbf{X}_t} p(\mathbf{X}_t | \mathbf{X}_1)$: unguided score function

Training of noise predictors

The two score functions in (7) can be approximated by the noise predictors as:

$$p(\mathbf{x}_t|y, \mathbf{x}_1) \approx -\frac{\hat{\epsilon}_t^\theta(\mathbf{x}_t|y)}{\beta_t}$$
$$p(\mathbf{x}_t|\mathbf{x}_1) \approx -\frac{\hat{\epsilon}_t^\theta(\mathbf{x}_t|\oslash)}{\beta_t}$$

where we use $\oslash$ to denote "nothing".

Training of noise predictor (cont.)

- We will train both $\hat{\epsilon}_t^\theta(\mathbf{x}_t|y)$ and $\hat{\epsilon}_t^\theta(\mathbf{x}_t|\oslash)$ in an unified manner.

$$\begin{aligned} & \arg \min_{\theta} \mathbb{E}_{\square} \left[\left\| \hat{\epsilon}_t^\theta(\mathbf{x}_t|y) - \epsilon \right\|^2 \right] \\ &= \arg \min_{\theta} \mathbb{E}_{\square} \left[\left\| \hat{\epsilon}_t^\theta(\alpha_t \mathbf{x} + \beta_t \epsilon | y) - \epsilon \right\|^2 \right] \end{aligned}$$

where

$$\square = t \sim \mathcal{U}[0, 1], \quad (\mathbf{x}, y) \sim p_{data}, \quad \text{with probability } \eta, y \leftarrow \oslash, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$$

Remark: In practice, it is one neural network $\hat{\epsilon}_t^\theta(\mathbf{x}_t|y)$ that includes both the guided and unguided noise predictors.

Sampling

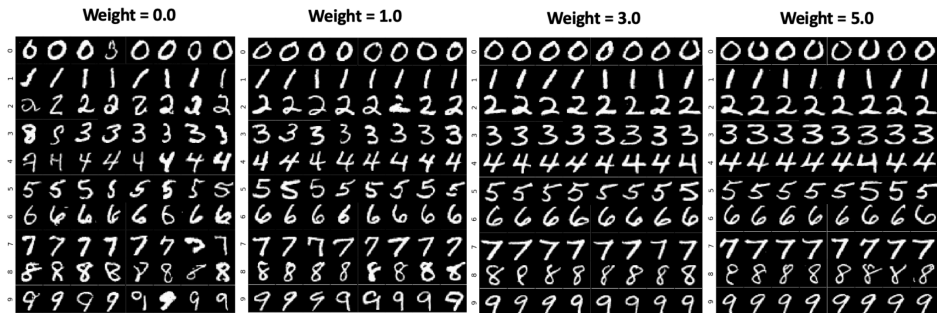
Algorithm 1 Classifier-free guidance sampling

- 1: Input: neural network $\hat{\epsilon}_t^\theta$, text prompt y , guidance w , number of steps k
 - 2: Set $t = 0$
 - 3: Set stepsize $h = \frac{1}{k}$
 - 4: Draw a sample $\mathbf{x}_0 \sim p_{init} = \mathcal{N}(0, \mathbf{I})$
 - 5: **for** $i = 1, \dots, k$ **do**
 - 6: $\hat{\epsilon}_t = (1 - w)\hat{\epsilon}_t^\theta(\mathbf{x}_t|\oslash) + w\hat{\epsilon}_t^\theta(\mathbf{x}_t|y)$
 - 7: $\mathbf{x}_{t+h} = \beta_{t+h}\hat{\epsilon}_t + \alpha_{t+h}\frac{(\mathbf{x}_t - \beta_t\hat{\epsilon}_t)}{\alpha_t}$ [DDIM sampling procedure]
 - 8: Set $t \leftarrow t + h$
 - 9: **end for**
 - 10: return $\mathbf{x}_{t=1}$
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Remark 1: In principle, any ODE solver can be applied to perform sampling for a diffusion model.

Demonstration of classifier-free guidance

- For very high guidance weight, very sharp images are generated that look very realistic. However, they are not that diverse. In a way, they all look the same.
- For very low guidance weight, we see a large variety in the samples per class. However, they are partially of very low quality and they are quite blurry.



References

- Lecture notes by MIT
- Tutorial on Flow Matching for Generative Modeling
- Introduction to Brownian Motion

Questions?